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Epistemic diversity across language models mitigates knowledge collapse

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As artificial intelligence (AI) becomes more widely used, concerns are growing that model collapse could lead to knowledge collapse, i.e. a degradation to a narrow and inaccurate set of ideas. Prior work has demonstrated *single*-model collapse, defined as performance decay in an AI model trained on its own outputs. Inspired by ecology, we ask whether increasing AI ecosystem diversity (i.e., the number of distinct models) can mitigate such collapse. To study the effect of diversity on model performance, we extend the single-model approach by segmenting the training data across an increasing number of language models and evaluating the resulting *ecosystems* of models over ten self-training iterations. We find that training a single model on the entire dataset improves performance only in the short term but amplifies collapse over longer horizons. Specifically, we observe that the *optimal* diversity level (i.e., the level that maximizes performance) increases monotonically with the number of self-training iterations. The observed effect is robust across various experimental settings, including different model families, parameter sizes, mixing human- and model-generated data, and temperature sampling methods, demonstrating the significance of ecosystem diversity for mitigating collapse. Moreover, our experiments with increased model and dataset sizes indicate that scaling up the system can amplify collapse in highly homogeneous ecosystems, thereby increasing the diversity benefits. In light of AI monoculture concerns, our results suggest the need to monitor (dis)agreement among AI systems and to incentivize more domain- and community-specific models to ensure successful knowledge production in the long run.

Additional Key Words and Phrases: artificial intelligence, epistemic diversity, knowledge collapse, model collapse, AI ecosystem, system pluralism, AI pluralism

1 Introduction

The growing role of Artificial Intelligence (AI) in knowledge dissemination and curation raises concerns about a collapse in human knowledge [49, 59, 64]. The scenario unfolds as follows (Figure 1): as AI systems become widely incorporated in private and professional contexts (e.g., in scientific discovery and writing [30, 41, 46]), they also generate an increasing volume of artificial data used for training. This amplifies the risk of model collapse, a degenerative process in which generative models re-trained on their own outputs begin producing homogeneous, biased, and nonsensical outputs [8, 31, 56]. Ultimately, such homogenization in model-generated data, particularly with language, may contribute to knowledge collapse, a degradation to a narrow, and potentially inaccurate, set of ideas [46, 49, 68].

Independent of these recent AI-related concerns, philosophers have argued for the benefits of epistemic diversity in science, not only for advancing knowledge production, but also to prevent injustices such as the undervaluing of minority perspectives [18, 24, 43, 52, 57]. When scientists learn from both their own results and one another's findings, diversity reduces the risk of converging on false beliefs. For example, monoculture thinking in medicine prolonged the myth that sickle cell disease was limited to Black Americans, leading not only to discrimination against the Black community but also to reduced treatment options for anyone affected by the disease. [4].

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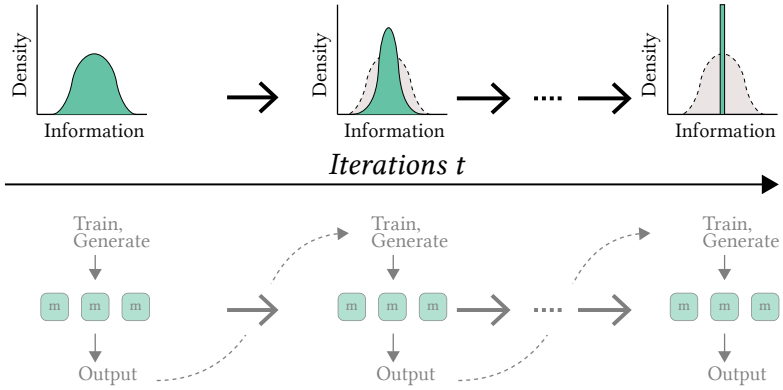

Fig. 1. **Illustration of model collapse in AI ecosystems.** Because of a model's (m) inherent biases (in learning and resampling) and finite data, it cannot perfectly regenerate the information space (e.g., concepts, claims, facts) it was trained on. Some information becomes distorted, and some is lost entirely due to low sampling probabilities. When this training–generation cycle is repeated, the model's output becomes increasingly homogeneous and degrades over iterations t. The probability distribution of the information space expressed in the output (and represented by the model, p_θ) gradually narrows with increasing t, converging toward delta-like functions. Collapse can occur at both the individual model level and the ecosystem level, meaning that even a collection of models trained on their collective output can eventually degenerate (here illustrated with three models). In society, we expect to advance knowledge. Without human intervention, the best we can hope for from AI models, though, is that they preserve the original distribution of information. Thus, for the purpose of this paper, we evaluate ecosystems based on model performance on a fixed ground-truth dataset over 10 self-training iterations.

"If diversity is good, how much diversity is best?" [57]. Simulations of collective knowledge production identify an upper bound on diversity: once diversity increases to the point where scientists no longer learn from one another, overall success rates begin to decline[1] [20, 50, 69, 75].

Like scientists exchanging new findings, AI models learn from their own output and the output of other AI[2]. In the presence of AI monoculture—where a few large models, trained on similar datasets with similar model architectures, dominate the AI ecosystem [10, 40, 44, 59, 71]—and where evidence of model collapse exists, we ask whether increasing ecosystem diversity (i.e., the number of dissimilar models) could mitigate monocultural shifts, and whether this happens monotonically or to an optimal degree.

In a homogeneous ecosystem, collapse occurs because of a model's non-zero probability of losing information or learning wrong patterns, which is irreversible, when relying only on its own output [56]. Analogous to productive disagreement among humans, we hypothesize that dissimilar models trained on their collective outputs can improve collective performance by correcting each other's errors. Viewing an AI model as an epistemological representation, we use the Hill-Shannon Diversity (HSD, D) from biology [32, 36, 53] to measure ecosystem diversity, which quantifies the effective number of distinct AI models. It is important to note that *the diversity across models serves as the primary independent variable that we manipulate to examine its effect on performance.* This

[1]To clarify our position: We do not believe this upper bound has ever been reached.

[2]We are not attempting to compare humans and AI models underlying learning mechanisms but only to illustrate information interactions in human and AI ecosystems. We use terms like "learning" and "training" because they are established in AI research but recognize the problems in anthropomorphizing this technology.

stands in contrast to prior work, which has focused on diversity in data [e.g., 55, 74] and/or has treated diversity as an outcome variable [e.g., 68].

Given the enormous resources required to train and operate prevailing models (e.g., ChatGPT, Gemini, Claude, etc.), studying the impact of diversity on AI ecosystem evolution at real-world scale is infeasible. And, even if we had infinite resources, many confounding factors could shape model performance, including model and data size [8, 37], model architecture, and system design. To address these challenges, we build upon previous work that analyzes a simple, single-model in a semi-controlled environment [e.g. 31, 56]. This previous work re-trains small language models on their own output over multiple iterations and measures performance using perplexity. We take a similar approach, but instead focus on ecosystems of models (i.e., collections of models), rather than a single model. This allows us to study the effects of ecosystem diversity on model performance, which we measure separately on two distinct model families (OPT [72] and GPT2 [51]).

While models can be differentiated in many ways, (e.g., by size, architecture, alignment, system messaging)[3], *we instantiate ecosystem diversity through variation in training data.*. To do this under a controlled setting, we segment the training data, which is fixed in size and, in the first iteration step, also in content. Through this process, we go from one, pre-trained model to multiple fine-tuned models (M=1, 2, 4, 16), as shown in Figure 2. We use Wikitext2 [45] as the dataset for generating artificial training data, evaluation, and initial training. This segmentation approach allows us to vary ecosystem diversity ($D = M$) in a relatively simple manner, while keeping factors, such as model family and the per-ecosystem resource budget for sample size and computation, fixed. In our primary experimental setting, we use the smallest versions of the two model families (OPT-125m and GPT2).

Our experiments identify an optimal level of ecosystem diversity for model performance, which increases monotonically with the number of self-training iterations (Figure 3). In other words, the prevailing strategy of training a single model on as much data as possible is short-sighted: over longer horizons, a collection of models trained on segmented datasets outperforms a single model. As the number of self-training loops increases, stronger segmentation is required to reach the diversity level that maximizes overall performance. Put simply, the benefits of diversity grow with the number of iterations.

To assess the effect of ecosystem diversity under more realistic conditions, we introduce two variations of our primary setting. In the first variation (V1), we increase the size of system (i.e., model and data size). Scaling up leads to a stronger collapse in more homogeneous ecosystems, while collapse is attenuated in more diverse settings, resulting in increased diversity benefits. The second variation (V2) enhances the training data by (1) incorporating 10% fresh, original Wikitext2 data at each iteration and (2) by applying temperature sampling with $\tau = 0.5, 1, 2$. Prior work suggests that both real-data incorporation [28, 56, 70] and data diversification [55, 68, 74] can mitigate collapse. While our results confirm these findings, *they also show that the effect of ecosystem diversity outweighs the effects of data quality manipulation.* Overall, the experimental variations suggest that the benefits of diversity for long-term performance may persist under real-world AI ecosystem conditions, where models such as ChatGPT are substantially larger, trained on far more data, can be optimized for diverse outputs, and where human-generated data is not replaced but instead accumulates alongside artificial data.

[3]A specific model architecture can be used to build different AI systems (i.e., the end products with which users interact) [11]. For simplicity, we use "model family" to differentiate between OPT and GPT2 pre-trained base models and "model" to refer to the fine-tuned versions.